

Pay-Equit with Supe Learning

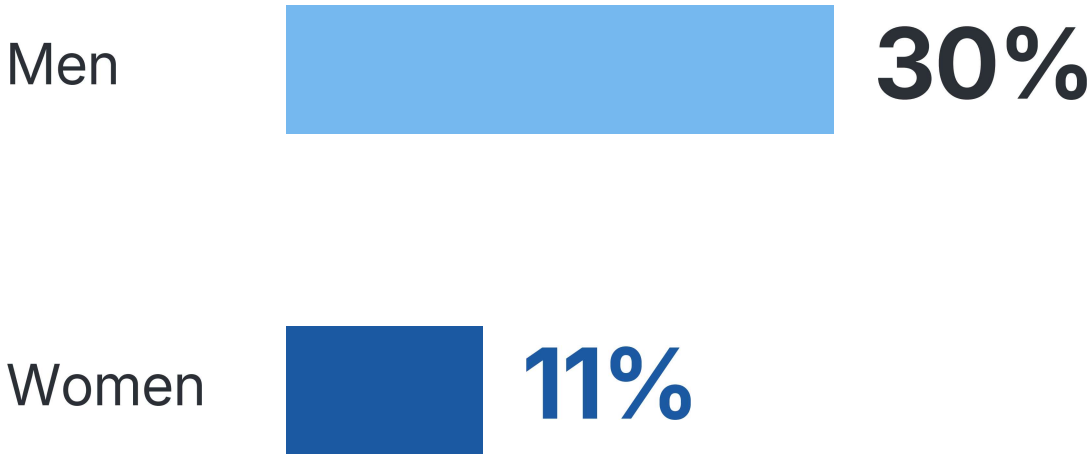
Our role: a pay-equity regulator

Is the pay gap explained by the job?

Our questions: Which employers should we investigate? Does the gap

All workers

Share earning more than $50K



Same de

Share earnir

Men

Women

Our data and what the model may u

USED · job-related factors

Age

Education (years)

Occupation

Hours per week

Workclass

Investment income

EXCLU

~~Sex~~

~~Race~~

~~Nativ~~

~~Relat~~

~~Marit~~

Husb
relat

One pipeline, three models

Raw data

Feature engineering
+5 new features

Average precision on the validation set (main metric: focuses on the rare >$50K class)

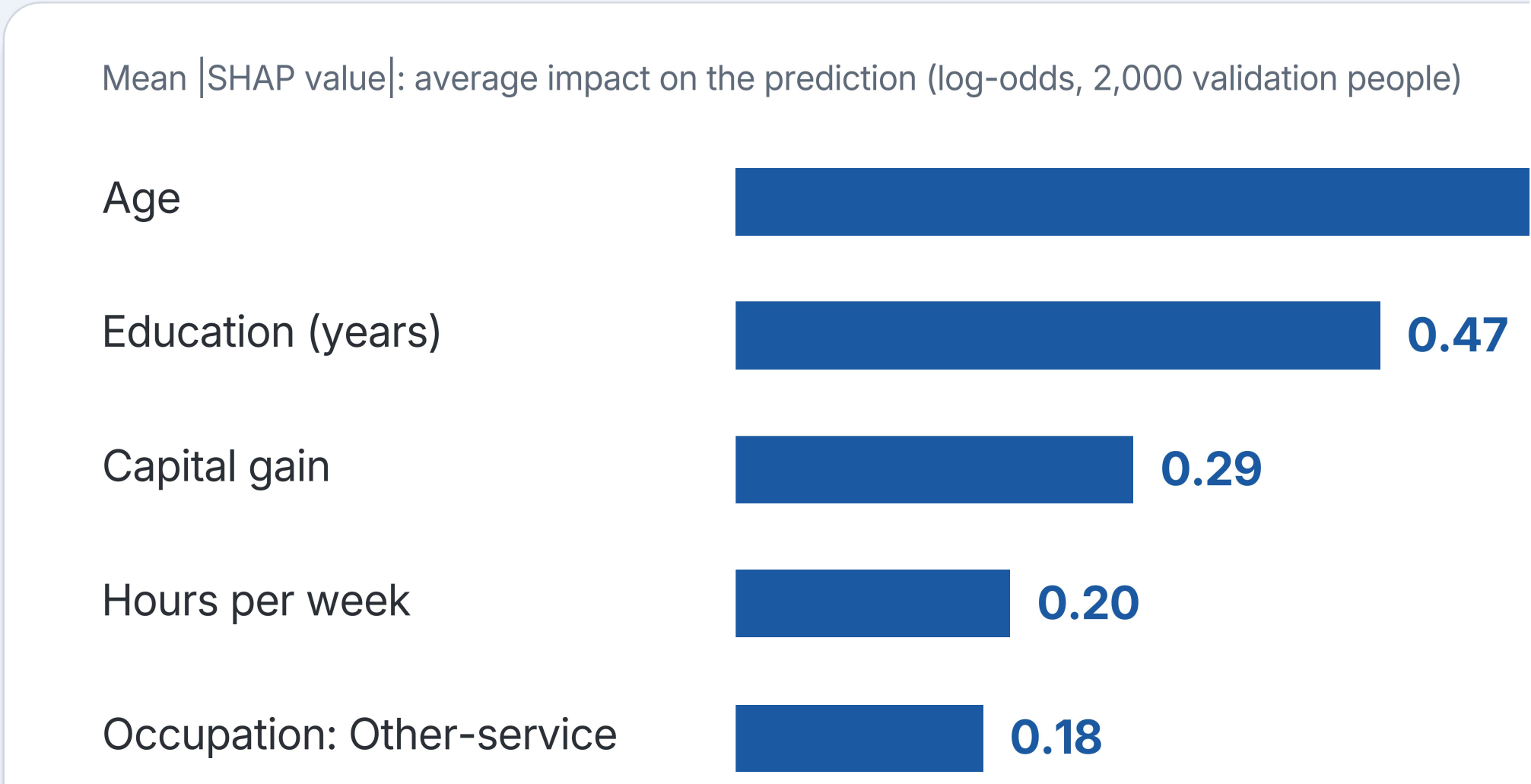
Logistic Regression

0.692

Random Forest

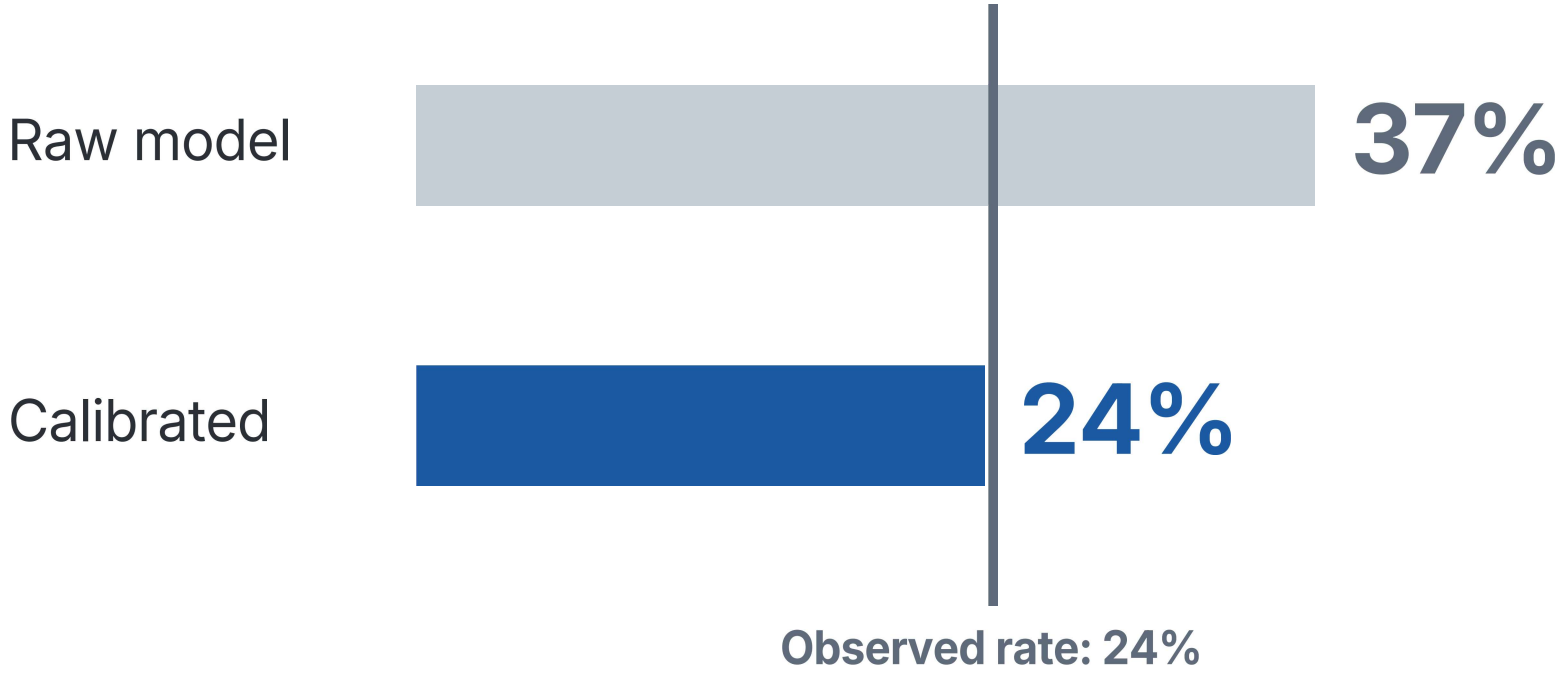
0.752

What drives the predictions (SHAP)



Can we trust the probabilities?

Mean predicted probability vs observed rate



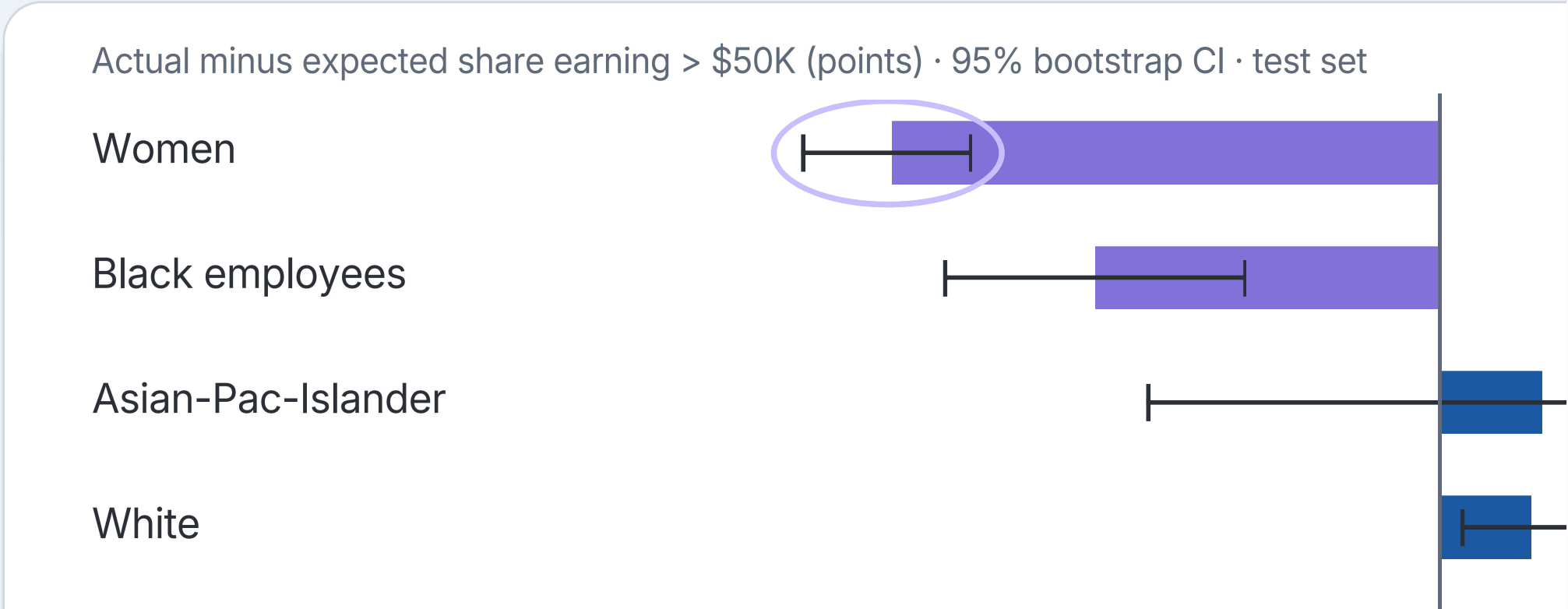
Cal

raw
0.01

The unexplained gap

−5.4 p

Women vs. what their job-related prof



Why not use all columns?

Before: all columns

incl. sex, race, relationship, marital status

ROC-AUC

0.930

Unexplained gap, women

+0.2 pts

Afte

no sex

ROC-

0

Unexp

From model to tool

FastAPI service

POST /audit

An inspector sends a company's employee records → gap per group, 95% CI, flagged employees

```
{ "women": { "gap": -6.0, "ci": [-7.8, -4.0] },
  "flagged_underpaid": 52 }
```

Simulated company of 2,000 employees

POST /predict

One profile → expected probability. Same profile, any sex or race: same answer.



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Thank you for listening